

Chapter 7. Quantum Search and Quantum Counting ①

Problem:

Input: $n \in \mathbb{N}$ and a black-box that implements a function

$$f: \{0, 1\}^n \rightarrow \{0, 1\}.$$

$$|\{0, 1\}^n| = 2^n = N.$$

$$\text{if } |\{x: f(x)=1\}| = M.$$

Output: A string $\vec{x} \in \{0, 1\}^n$ with $f(\vec{x}) = 1$. then.

$$\text{Probability: } \frac{M}{N}.$$

★ Grover's search algorithm $\rightarrow$ quadratic speedup. $\overset{\text{classic}}{O(N)} \leftarrow \overset{\text{search}}{N-M+1} \text{ times (worst case)} \rightarrow$ to get the object.

Applications: unstructured search / brute force. $O(\sqrt{N})$

Cryptography (key search / hash preimages)

- ① Grover search algorithm with a known number of solutions.
- ② ~~Las Vegas~~ Las Vegas algorithm with a unknown number of solutions.
- ③ Quantum Counting.

7.1.2.

"Measure the quantum state"

$$P(f(\vec{x})=1) = \sin^2 \theta = \frac{M}{N}.$$

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{\vec{x} \in \{0, 1\}^n} |\vec{x}\rangle = \sqrt{\frac{N-M}{N}} |s_0\rangle + \sqrt{\frac{M}{N}} |s_1\rangle = \cos \theta |s_0\rangle + \sin \theta |s_1\rangle.$$

$$\text{with } |s_0\rangle = \frac{1}{\sqrt{N-M}} \sum_{\substack{\vec{x} \in \{0, 1\}^n \\ f(\vec{x})=0}} |\vec{x}\rangle, \quad |s_1\rangle = \frac{1}{\sqrt{M}} \sum_{\substack{\vec{x} \in \{0, 1\}^n \\ f(\vec{x})=1}} |\vec{x}\rangle.$$

$$\sin \theta = \sqrt{\frac{M}{N}} \Leftrightarrow \theta = \arcsin \sqrt{\frac{M}{N}}.$$

"Amplitude amplification": increase the amplitude of state $|s_1\rangle$.

"G"

$$G(\cos \alpha |s_0\rangle + \sin \alpha |s_1\rangle) = \cos(\alpha + 2\theta) |s_0\rangle + \sin(\alpha + 2\theta) |s_1\rangle.$$

Grover iterator
(unitary operator)

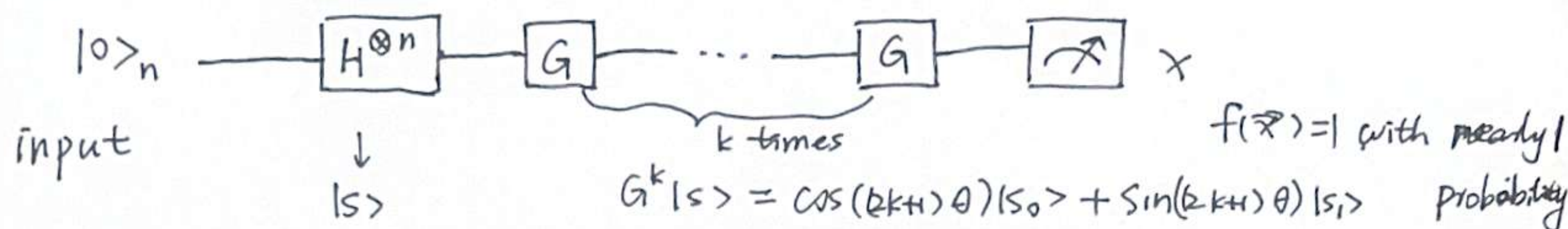
$$\text{then } G(|s\rangle) = \cos(3\theta) |s_0\rangle + \sin(3\theta) |s_1\rangle.$$

$$\Rightarrow G^k(|s\rangle) = \cos((2k+1)\theta) |s_0\rangle + \sin((2k+1)\theta) |s_1\rangle.$$

$$k \nearrow \left\lfloor \frac{\frac{\pi}{4\theta}}{2} - \frac{1}{2} \right\rfloor.$$

"Quantum circuit for Grover's search algorithm."

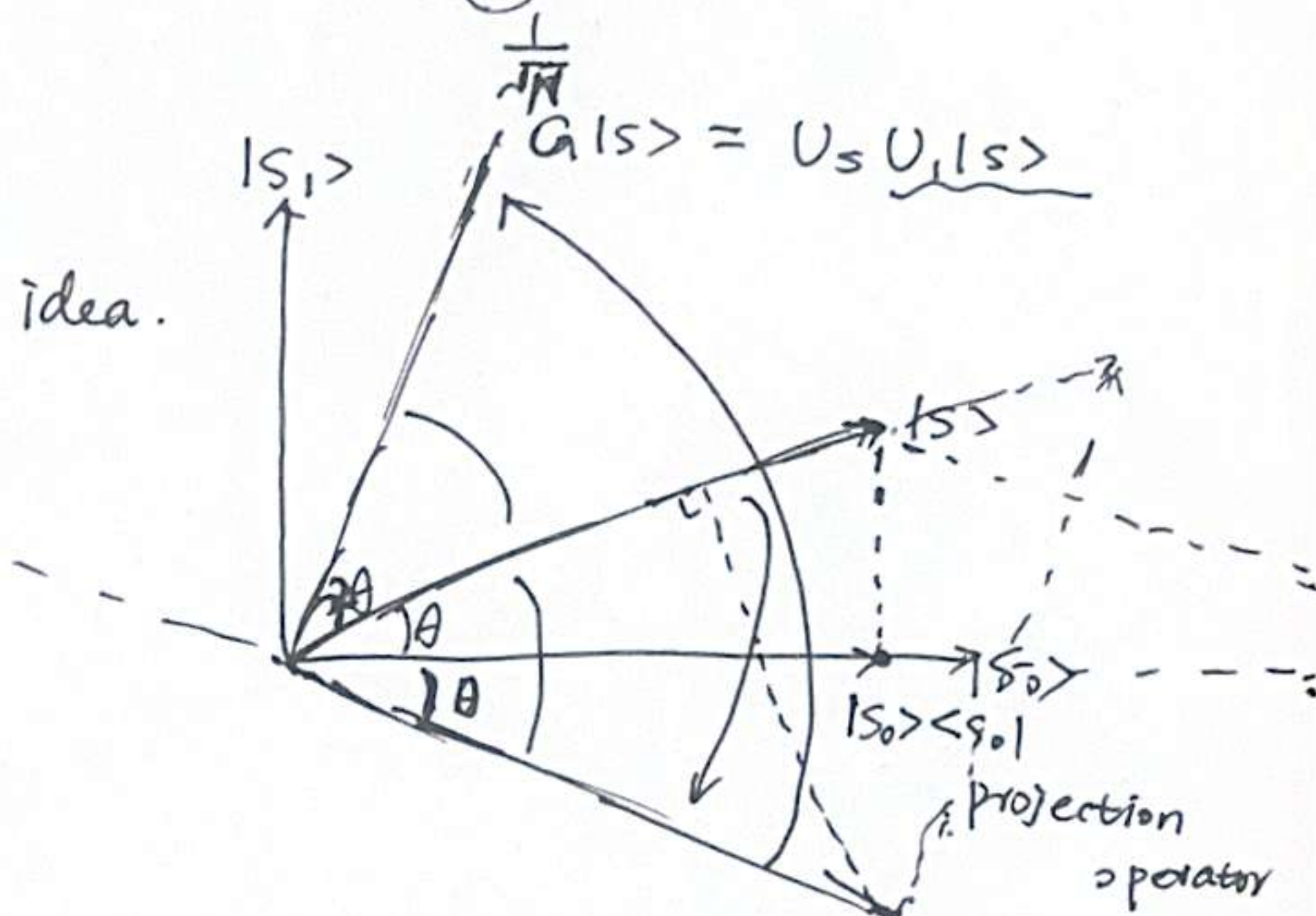
(2)



$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \Rightarrow |s\rangle = H^{\otimes n} |0\rangle_n = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle_n$$

"The Grover iterator"

- ① oracle U_1
- ② diffusion operator U_s



$$U = U_s U_1$$

$$= (2|s\rangle\langle s| - I)(I - 2|s_1\rangle\langle s_1|) \Rightarrow U_1 = 2|s_0\rangle\langle s_0| - I = I - 2|s_1\rangle\langle s_1|$$

$$\text{then } U_1 |s\rangle = 2|s_0\rangle\langle s_0| |s\rangle - |s\rangle$$

$$2^\circ. U_s = 2|s\rangle\langle s| - I$$

~~How to~~

"Implementation of the Grover iterator"

For oracle U_1

$$U_1 |s\rangle = U_1 (\cos\theta |s_0\rangle + \sin\theta |s_1\rangle)$$

$$= \cos\theta |s_0\rangle - \sin\theta |s_1\rangle$$

$$(-1)^0 \cos\theta |s_0\rangle + (-1)^1 \sin\theta |s_1\rangle$$

$$(a \vee b) \wedge \neg(a \wedge b)$$

P207. (5.1.2)

$$U_f : H_2 \rightarrow H_2 \quad |x\rangle|y\rangle \mapsto |x\rangle|f(x) \oplus y\rangle = |x\rangle X^{f(x)} |y\rangle$$

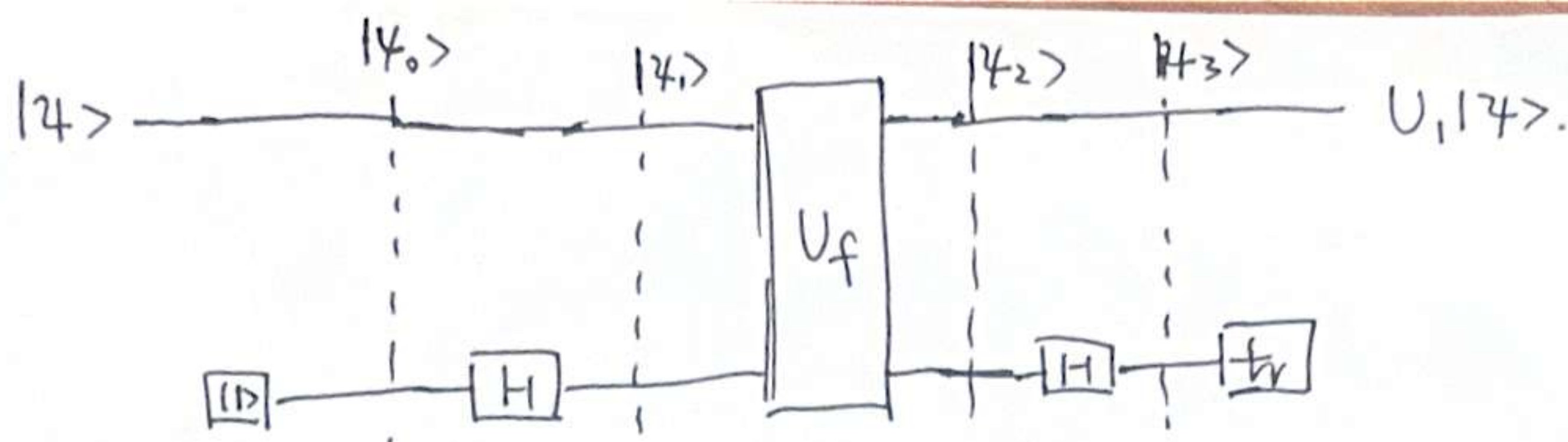
" $\oplus$ " XOR.

$$U_f |x\rangle|1\rangle = |x\rangle|f(x) \oplus 1\rangle = \begin{cases} \text{if } f(x)=0. & = |x\rangle|0 \oplus 1\rangle = |x\rangle|1\rangle \\ \text{if } f(x)=1. & = |x\rangle|1 \oplus 1\rangle = |x\rangle|0\rangle \end{cases}$$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

flip.

$$U_f |x\rangle|0\rangle = |x\rangle|f(x) \oplus 0\rangle = \begin{cases} |x\rangle|0 \oplus 0\rangle = |x\rangle|0\rangle & f(x)=0. \\ |x\rangle|1 \oplus 0\rangle = |x\rangle|1\rangle & f(x)=1. \end{cases}$$



(3)

$$|\psi\rangle = \cos\alpha |s_0\rangle + \sin\alpha |s_1\rangle.$$

$$|\psi_0\rangle = |\psi\rangle |1\rangle = \cos\alpha |s_0\rangle |1\rangle + \sin\alpha |s_1\rangle |1\rangle$$

$$|\psi_1\rangle = |\psi\rangle H|1\rangle = |\psi\rangle \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$|\psi_2\rangle = U_f |\psi_1\rangle = U_f |\psi\rangle \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$= \frac{1}{\sqrt{2}} (U_f |\psi\rangle |0\rangle - U_f |\psi\rangle |1\rangle)$$

$$= \frac{1}{\sqrt{2}} (\cos\alpha U_f |s_0\rangle |0\rangle + \sin\alpha U_f |s_1\rangle |1\rangle - \cos\alpha U_f |s_0\rangle |1\rangle - \sin\alpha U_f |s_1\rangle |0\rangle)$$

$$= \frac{1}{\sqrt{2}} (\cos\alpha |s_0\rangle |0\rangle + \sin\alpha |s_1\rangle |1\rangle - \cos\alpha |s_0\rangle |1\rangle - \sin\alpha |s_1\rangle |0\rangle)$$

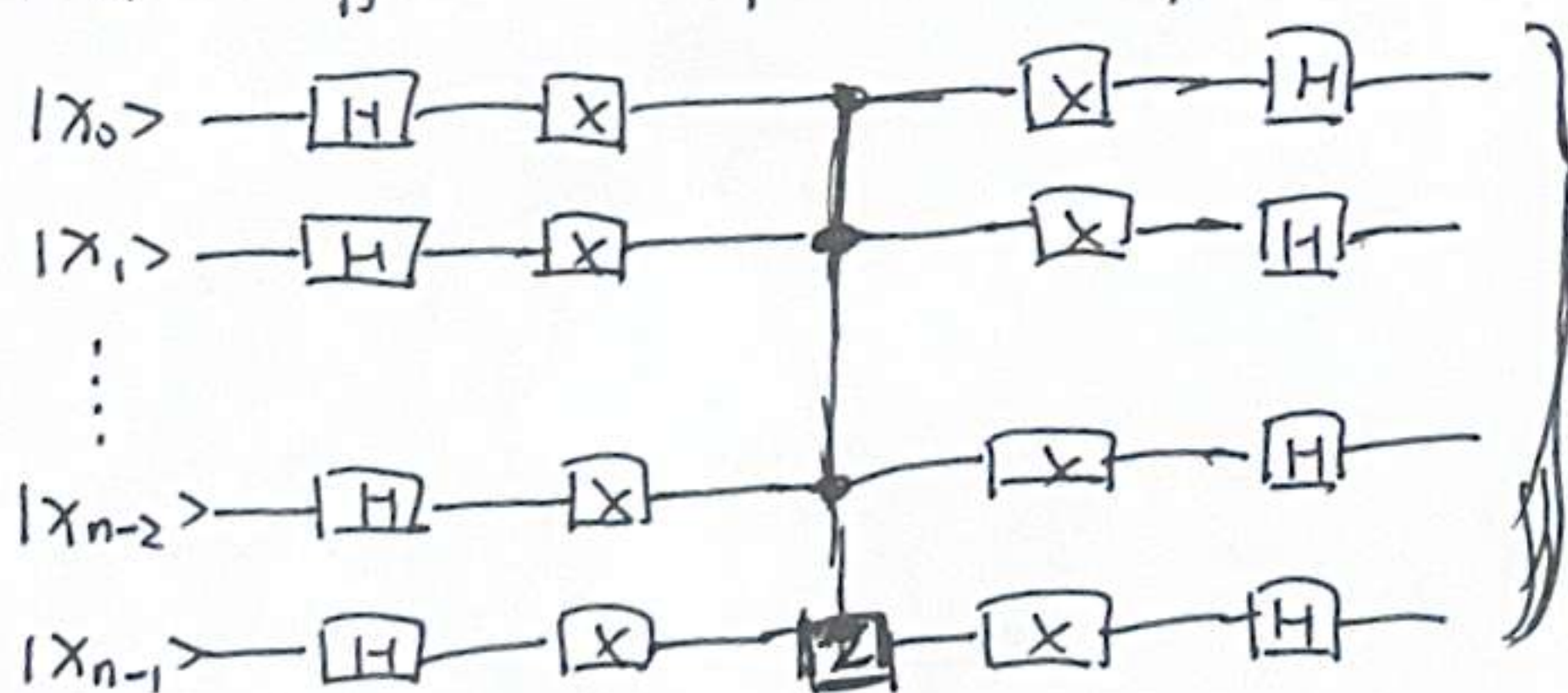
$$H|x\rangle = |1\rangle$$

$$= \cos\alpha |s_0\rangle |x\rangle - \sin\alpha |s_1\rangle |x\rangle$$

$$\Rightarrow |\psi_3\rangle = \cos\alpha |s_0\rangle H|x\rangle - \sin\alpha |s_1\rangle H|x\rangle = (\cos\alpha |s_0\rangle - \sin\alpha |s_1\rangle) |1\rangle$$

$$\Rightarrow U|\psi\rangle = \cos\alpha |s_0\rangle - \sin\alpha |s_1\rangle.$$

For diffusion operator $-U_s$.



$O(n)$ elementary quantum gates.

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$Z|0\rangle = |0\rangle \\ Z|1\rangle = -|1\rangle$$

$$-U_s = H^{\otimes n} R_0 H^{\otimes n} = -(2|s\rangle\langle s| - I)$$

$$\text{with } R_0 = I - 2|0\dots 0\rangle\langle 0\dots 0|.$$

$$\text{Let's show that } V = 2|0\rangle_n\langle 0|_n - I = -X^{\otimes n} C^{n+1}(Z) X^{\otimes n}.$$

Let $\vec{x} \in \{0, 1\}^n$, then

(4)

$$\forall |\vec{x}\rangle = (2|0\rangle_n \langle 0|_n - \mathbb{I}) |\vec{x}\rangle = \begin{cases} |\vec{x}\rangle, & \text{if } \vec{x} = \vec{0} \\ -|\vec{x}\rangle, & \text{if } \vec{x} \neq \vec{0} \end{cases}$$

Since $X^{\otimes n} |\vec{x}\rangle = |\neg \vec{x}\rangle, \Rightarrow$

$$C^{n-1}(Z) X^{\otimes n} |\vec{x}\rangle = \underbrace{C^{n-1}(Z)}_{|1 \dots 1\rangle \text{ flip.}} |\neg \vec{x}\rangle = \begin{cases} -|\neg \vec{x}\rangle, & \text{if } \vec{x} = \vec{0} \\ |\neg \vec{x}\rangle, & \text{if } \vec{x} \neq \vec{0} \end{cases}$$

~~or~~ otherwise. no.

$$\Rightarrow X^{\otimes n} C^{n-1}(Z) X^{\otimes n} |\vec{x}\rangle = \begin{cases} -|\vec{x}\rangle, & \text{if } \vec{x} = \vec{0} \\ |\vec{x}\rangle, & \text{if } \vec{x} \neq \vec{0} \end{cases}$$

$$\Rightarrow V = -X^{\otimes n} C^{n-1}(Z) X^{\otimes n}$$

2°. check. $U_S = H^{\otimes n} (2|0 \dots 0\rangle \langle 0 \dots 0| - \mathbb{I}) H^{\otimes n}$

$$= 2H^{\otimes n} |0 \dots 0\rangle \langle 0 \dots 0| H^{\otimes n} - H^{\otimes n} \cdot H^{\otimes n}$$

$$= 2|S\rangle \langle S| - \mathbb{I}$$

"Analysis"

Theorem 7.1.21.

$n \in \mathbb{N}$. $N = 2^n$. $f: \{0, 1\}^n \rightarrow \{0, 1\}$. $M = |f^{-1}(1)| > 0$.

return a $\vec{x} \in \{0, 1\}^n$, s.t. $f(\vec{x}) = 1$, with probability ~~$\frac{1}{N}$~~ $\sin^2((2k+1)\theta)$

k times $G \approx k(U_f) + k \cdot \frac{O(n)}{O(\log N)}$

$$> \sin^2 \theta = \frac{M}{N}$$

proof: $(2k+1)\theta \leq \frac{\pi}{2}$

$$\Rightarrow k \leq \frac{\pi}{4\theta} - \frac{1}{2}$$

$$\Rightarrow k = \lfloor \frac{\pi}{4\theta} - \frac{1}{2} \rfloor$$